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MECH 4692 - RENEWABLE ENERGY

Final Report

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Executive Summary

Greenhouse gases (GHG) are a primary contributor to climate issues worldwide. These gases, such as carbon dioxide, are a result of burning fossil fuels and allowing the toxins to be emitted into the atmosphere. Since the main contributors to these emissions come from residential houses, this report shows the steps taken to retrofit a Winnipeg home by following the Renewable Energy Demand (RED) model and with a budget of \$150,000.

The first step that was implemented in this report was determining the demand of the house and what the current power sources are. This was done by taking the natural gas, hydro, gasoline, and propane annual usages and determining what the total demand of the house is. It was found that the total annual demand of the Winnipeg house being retrofitted is 1,200 MWh.

The second step that was implemented in this report was determining which of the current technology in the house need to be changed to improve the overall efficiency of the home. With all the devices used in the house, most were converted to their electrical counterparts whereas a biomass boiler and barbeque were implemented instead. For the heat pump, it was eliminated from the home where heating and cooling will instead be supplied from the heat panels and biomass boiler. Contrarily, for the insulation, they were not changed as the house was recently retrofitted with new insulation. Additionally, for the windows, they are all to be changed to triple-glazed windows (TGW) is to ensure that the house is insulated in the winters and repels heat in the summers. Furthermore, it was determined that the three vehicles currently being used in the home were to be sold allowing for the purchase of a Nissan Leaf Electric 4WD.

The third step that was implemented in this report was determining which renewable energy sources should be implemented to satisfy the demand requirements of the home. It was determined that on the house, heat panels will be implemented, whereas in the house, a heat recovery ventilator (HRV) will be used. Additionally, a photovoltaic (PV) array will be placed on the lawn adjacent to the home. Furthermore, to satisfy energy demands, two wind turbines and additional heat panels will be placed off-grid on a different property.

With all the changes and additions, the total RER value became 0% from the previous amount of 3.88%. Also, the total amount of money left over from the \$150,000 is \$18,826.29.

Contents

1	Introduction	1
2	Assumptions	1
3	Data Acquisition - Demand	3
3.1	Initial RER and GHG Emissions	3
3.2	Initial Costs	4
4	Proposed Design and Technologies	5
4.1	Technology Changes - Efficiency	5
4.2	Technology Additions - Renewable	6
4.3	Proposed Drawings of Implemented Devices	7
5	Model Description	7
5.1	Solar Panel Analysis	8
5.2	Turbine Analysis	10
5.3	Biomass Analysis	12
6	Power and Heat Balance	12
7	References	13
8	Appendix A	15
9	Appendix B	15
10	Appendix C	16
11	Appendix D	17
12	Appendix E	20
13	Appendix F	21
14	Appendix G	21
15	Appendix H	22
16	Appendix I	22

List of Figures

1	Load probability and electrical loads for a local residential hydro meter in Winnipeg, Manitoba showing the load probability (a) and the hourly distribution of electricity (b).	3
2	GHG emissions for a local residential home in Winnipeg, Manitoba showing the emissions for natural gas, gasoline, and propane. Additionally, the RER is shown.	4
3	On-House Heat Panels where 36 panels were used and a total annual production of 48.7 MWh.	9
4	Off-Grid Heat Panels where 44 panels were used and a total annual production of 59.5 MWh.	9
5	On-House PV Panels where 2 panels were used and a total annual production of 1.6 MWh.	9
6	Off-Grid PV Panels where 3 panels were used and a total annual production of 1.6 MWh.	9
7	Frequency of occurrence per month (a) and Cumulative frequency of occurrence per month (b) for the wind velocity approximation for a given year. . . .	10
8	Wind probability, showing both the individual probability, f_i , (a) and the summed probability, F_i , (b). Additionally, this data shows the values are satisfied as they converge to 1.0.	11
9	Turbine power for an average year showing the fluctuation of power generation where a total power of 178 kWh was produced.	11
10	Various power generation and usages. Also, the amount of power that is excess which will be sold back to Manitoba Hydro is shown.	12
11	Annual car usages for a 2015 Ford Escape, 2012 Ford Platinum, 2005 Chrysler Grand Caravan	15
12	Annual energy usages for a house	16
13	Annual GHG emissions and gasoline usages for a 2015 Ford Escape, 2012 Ford Platinum, 2005 Chrysler Grand Caravan	16
14	Current house	17
15	Concept house	18
16	Off Grid Concept	19
17	Bin# Calculations to find the probability curve for the wind turbine	20
18	Rayleigh approximation data	20
19	Weibull approximation data	21

20	Biomass data showing the overall emissions is very small and therefore can be assumed to be zero.	21
21	Power and heat balance showing the leftover power after usages per month and the total amount of money that is needed to be paid to Manitoba Hydro.	21
22	Power and heat balance showing the total amount saved in 30 years if all renewables were implemented.	22
23	Final RER analysis	22

List of Tables

I	Assumptions used in the project	2
II	Initial Annual Costs for natural gas, hydro, gasoline, hydro, and propane . . .	5
III	Costs for Technology Changes	6
IV	Costs for electric technology implementation	7
V	Recorded data from a local hydro meter in Winnipeg, Manitoba showing the meter reading, load, and load probability.	15
VI	Conversion values for natural gas, gasoline, and propane.	16

1 Introduction

As the world is moving to renewable energy as a means of power generation, leading GHG emitters such as homes, are requiring a retrofit. Whether it is natural gas, gasoline, or propane, 21st-century homes contribute to roughly 40% of the total GHG emissions yearly worldwide [1]. Therefore, to combat this issue, homeowners can eliminate or convert the emission sources from homes to renewable energy sources such as solar, wind, hydro, or geothermal. This report will outline the necessary steps required to retrofit a home to decrease the amount of GHG it emits while also showing the proposed design plan for an existing home in Winnipeg, Manitoba, Canada given a \$150,000 budget. Additionally, this report aims to show how the RED model is implemented to ensure a true net zero home.

2 Assumptions

To complete the analysis proposed in the Introduction Section, various assumptions were needed. These assumptions take into account the weather, vehicles, loads, energy alternatives, electric vehicles, and biomass analyses in the project. The different assumptions mentioned and their explanations can be seen in Table I below:

TABLE I. Assumptions used in the project

Section	Assumptions
Weather	The meteorological data used is from 2014 and will be sufficient for modeling the project in 2023.
Weather	The usage indicated for each vehicle is a rough estimate and will be taken as the actual annual usage.
Vehicles	There are more vehicles throughout the year that my family uses. We basically cycle through them. However, for this project, assume only the three recent vehicles are being used.
Vehicles	The 2015 Ford Escape uses 30% of the annual gasoline
Vehicles	The 2012 Ford Platinum uses 30% of the annual gasoline
Vehicles	The 2005 Chrysler Grand Caravan uses 30% of the annual gasoline
Loads	The amounts indicated represent a constant amount for every year
Loads	The lawnmower uses 3% of the annual gasoline
Loads	The Hedge Trimmer uses 1% of the annual gasoline
Loads	The snowblower uses 3% of the annual gasoline
Loads	For natural gas, the only appliances that use it in the house are the water heater, AC, and stove
Loads	The stove uses 30% of the annual natural gas
Loads	The AC uses 30% of the annual natural gas
Loads	The water heater uses 40% of the annual natural gas
Loads	For hydro, the only appliances that use it in the house are the lights and the outlets
Loads	The lights use 30% of the annual hydro
Loads	The outlets use 70% of the annual hydro
Loads	For the propane, the only appliance that uses it in the house is the barbeque
Loads	The barbeque uses 100% of the annual propane
Energy Alternatives	The PV array will be placed in the yard adjacent to the home
Energy Alternatives	The insulation in the house does not need to be changed as it was recently retrofitted
Energy Alternatives	The devices chosen are sufficient for the home being retrofitted
Electric Vehicle	The vehicle chosen is sufficient for the home
Biomass	The wood energy is going to heating the house in the winter and whenever the solar heating is not enough

3 Data Acquisition - Demand

The initial process for acquiring the necessary data for the load probability, daily electrical loads, and the yearly GHG emissions of a home consisted of reading a local hydro meter for a Winnipeg home. This data was recorded every hour for 24 hours, showing an increase in the total amount of energy used for a day. The measured data from a hydrometer can be seen in Appendix A where the total load demand in one day was found to be 32.1 kWh. Additionally, the total load demand annually was found to be 45 MWh.

Alongside the data acquired for the hydro, the natural gas, gasoline, and propane usages were also accumulated for a year. For the natural gas, the monthly house bills were accumulated and recorded on an Excel spreadsheet, which can be seen in Appendix B. Additionally, for the gasoline, the annual usage for a 2015 Ford Escape, 2012 Ford Platinum, 2005 Chrysler Grand Caravan, lawnmower, hedge trimmer, snow blower, chainsaw, and pressure washer was recorded using gasoline bills for a given year, which can be seen in Appendix C. Finally, for the propane, the annual usage was calculated based on the family barbeque, which is the only item that uses this energy source. This data can also be seen in Appendix B.

3.1 Initial RER and GHG Emissions

Once the data was obtained from the hydrometer, the load probability for one day was found. These values were found by taking the load values, which were obtained from the last dial on the hydrometer, and dividing them by the summed amount of all load values, which can be seen in Appendix A. To verify whether the data was correct, the load probability was summed and if it equaled 1, then the values were satisfied. A histogram of the values obtained in Appendix A can be seen below in Figure 1:

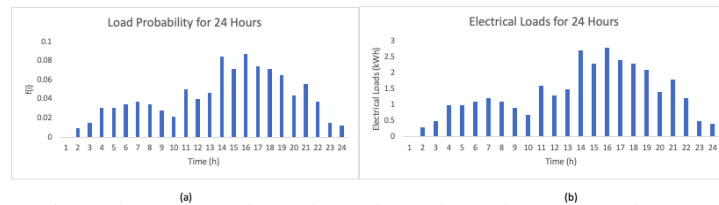


Figure 1. Load probability and electrical loads for a local residential hydro meter in Winnipeg, Manitoba showing the load probability (a) and the hourly distribution of electricity (b).

As seen in Figure 1, the histograms for the load probability and electrical loads for 24 hours are almost identical. This makes sense as the probability indicates at what time of the day the electrical load will be high or low. Therefore, when the probability increases, the electrical load increases, and vice-versa.

Once the load probability and electrical loads were obtained, the natural gas, gasoline, and propane usages were determined. These values were found by observing monthly natural gas bills in a home for a given year. Once these values were obtained, a VLOOKUP function was used on MS Excel to determine the hourly natural gas consumption for a year. By doing this, the total natural gas demand for a year was found to be 42 MWh. Additionally, the gasoline and propane annual demands were determined as discussed in the Data Acquisition Section to be 1,100 MWh and 100 kWh respectively. Following the accumulation of the annual usage for all non-renewable energy sources, the GHG emissions for each was found using the conversion table in Appendix C. These values can be seen in the pie chart below in Figure 2:

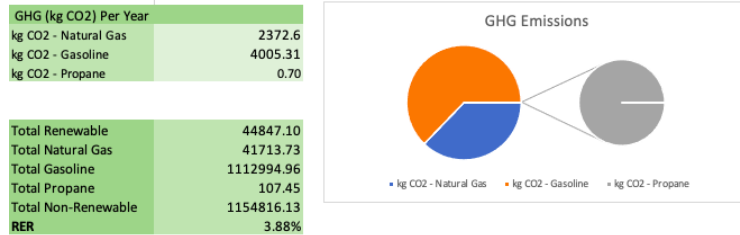


Figure 2. GHG emissions for a local residential home in Winnipeg, Manitoba showing the emissions for natural gas, gasoline, and propane. Additionally, the RER is shown.

Additionally, the renewable energy ratio (RER), which is a ratio that shows how much renewable energy is used in comparison to non-renewable energy, was found. The RER before any retrofits can be seen in Figure 2 as 3.88%. This value was found by dividing the total renewable energy usage, which in this stage is only hydro, by the non-renewable energy usage, which is natural gas, gasoline, and propane. Therefore, with the data seen in Figure 2, the total demand of the Winnipeg home being retrofitted can be found to be 1,200 MWh by summing the each usage.

3.2 Initial Costs

When considering the initial costs of project, the current hydro, gasoline, natural gas, and propane bills need to be accounted for. For a typical year in Winnipeg Manitoba, the annual

costs for the house that is being retrofitted in this project can be seen in Table II below:

TABLE II. Initial Annual Costs for natural gas, hydro, gasoline, hydro, and propane

Energy Source	Cost (\$)
Natural Gas	4,181.50
Gasoline	16,273.53
Hydro	1,709.16
Propane	64.95
Total	22,229.08

4 Proposed Design and Technologies

In order to reduce the amount of GHG emissions shown in Figure 2, various components in the home need to be changed alongside adding renewable sources for energy. This section will introduce the Renewable and Efficiency parts of the RED model by going over the proposed design concept that will highlight the changes and the new technology additions, either on or off the property.

4.1 Technology Changes - Efficiency

Currently, the home that is being retrofitted includes a Natural Gas Furnace, Natural Gas Stove, Heat Pump (for both heating and cooling), Gasoline Lawnmower, Gasoline Weed Eater, Gasoline Snow Blower, Gasoline Chainsaw, Gasoline Pressure Washer, Propane Barbeque, and three Gasoline Vehicles.

For each of the items listed above, the main sources of energy are hydroelectricity, natural gas, gasoline, and propane. Therefore, an electric, biomass, and thermal alternative will be used to replace them. Additionally, moving forward, a heat pump will no longer be required to heat or cool the home. Alternatively, solar heating and cooling will be used, which will be discussed in the next section, alongside heating and cooling from the biomass boiler.

For the insulation in the house, since there were recent retrofits done, there is no need to update the insulation as it is already at recommended R-ratings. Contrarily, for the windows, they will all be changed to TGW. This is to ensure that the house is insulated in the winters and repels heat in the summers. Furthermore, since the vehicles currently be-

ing used in the house are gasoline-powered, a Nissan Leaf Electric 4WD will be purchased [2].

The price and energy demand/produced of each of these changes can be seen in Table III below:

TABLE III. Costs for Technology Changes

Technology	Equipment Cost (\$)	Energy Required (kWh)	Energy Produced (kWh)
Biomass Boiler [3], [4]	24,487.71	0	4
Electric Stove [5]	1,195.00	6.7	0
Electric Lawnmower [6]	149.99	3.5	0
Electric Weed Eater [7]	79.99	0.218	0
Electric Snow Blower [8]	99.43	1.5	0
Electric Chainsaw [9]	199.99	3.5	0
Electric Pressure Washer [10]	149.99	2.4	0
Biomass Barbeque [11]	3499.99	0	0
10 24x36 TGW [12]	5550.00	0	0
10 36x48 TGW [12]	7500.00	0	0
4 12x24 TGW [12]	1200.00	0	0
Nissan Leaf Electric 4WD [2]	41,248.00	0.23	0
Level 2 Charger Port [13]	499.99	9.6	9.6
Total	85,630.08	28.63	13.6

4.2 Technology Additions - Renewable

Now that the changes have been made for the current home appliances, the energy sources need to be retrofitted alongside other additions. For the energy, there will be two sources: wind energy from turbines and solar energy from heat and PV panels. Additionally, an HRV will be included in the home to help make the air healthier and cleaner during the colder months. The price and energy required/produced of each of these additions can be seen in Table IV below:

TABLE IV. Costs for electric technology implementation

Technology	Amount	Equipment Cost (\$)	Energy Required (kWh)	Energy Produced (kWh)
Heat Panels [14]	80	29,068.0	0	25.6
PV Panels [15]	5	6,875.00	0	2.75
Panel Batteries [16]	7	9,651.39	22.5	0
Off-Grid Wind Turbines [17]	2	44,348.70	0	45
Turbine Batteries [16]	2	2,757.54	21	0
HRV [18], [19]	1	1,065.00	0.15	0
Total	90	107,965.63	14.145	58.35

Additionally, the total cost, energy required, and energy produced of the technology changed and additions are \$185,802.00, 31.96 kWh, and 62.35 kWh respectively.

In order to satisfy the \$150,000 budget, the vehicles indicated in the Data Acquisition Section will all be sold as they contribute to a majority of the GHG emissions. The amount for the sale of all three of the vehicles was found to be \$35,802.00 by using Kijiji and Facebook Marketplace. Assuming that these vehicles will be sold for the amount indicated, then this amount can be summed with the \$150,000 budget and be used to purchase an electric car. Therefore, the car that will be purchased will be a Nissan Leaf Electric 4WD [2]. Therefore, with this addition, the total cost left over will be \$18,826.29.

4.3 Proposed Drawings of Implemented Devices

To implement the devices in Sections 2.1 and 2.2, a drawing must be made that shows where each device will be placed on the house. For the purposes of this report, the heat and PV panels, biomass boiler, HRV, and stove are shown to be on or in the house, whereas the PV panels and turbines are off grid. These drawings can be seen in Appendix D.

5 Model Description

Once the current technology in the home was converted to their renewable alternatives and new energy generation devices were purchased, the actual power produced needs to be analyzed. This section will introduce the solar power, turbine, and biomass analyses used to complete the project.

5.1 Solar Panel Analysis

For the solar panels, it was found in Section 4.2 that the total energy produced is 28.35 kWh. However, this is the top, optimal power generation when the solar panels are working in ideal conditions. Therefore, this section will cover the analysis done to find the power produced by the on-house and off-grid heat panels, and the on-house and off-grid PV panels given the 2014 meteorological data.

Since the meteorological data analysis was used and not the clear sky model, the normal insolation radiation (I_{bn}) and the horizontal diffusion radiation (I_{dh}) were provided.

To start the analysis, an Excel spreadsheet was used to list every hour, day, and month for a year. Then, using the analysis process taught in MECH 4692, the following were calculated:

1. Solar angle (α) with calculating the solar time (t) based on the day-of-year ratio (B) and Equation of time (EOT) for every hour of the year.
2. Declination angle (δ) for every hour of the year.
3. Sun Zenith angle (χ) given the δ , latitude, and α .
4. Sun Azimuth angle (ζ) given the δ , latitude, and α .

Once the above values were found, the tilted normal insolation radiation ($I_{bn_{tilt}}$) and the titled horizontal diffusion radiation ($I_{dh_{tilt}}$) were calculated as the solar panels are tilted at an angle. Additionally, the horizontal insolation (I_h) and the tilted insolation (I_r) were also found for every hour of the year. After all the tilted insolation values were found, the total tilted insolation (I_{tilt}) was calculated by taking the sum of $I_{bn_{tilt}}$, $I_{dh_{tilt}}$, and I_r for every hour of the year.

Finally, the power generated every hour for the heat panels was calculated using formulas for the 10002114 solar panel data sheet found in [20]. Whereas for the PV panels, Equation 1 was used to find the power for every hour.

$$P_{PV} = \frac{I_{tilt} PV_{rating} N_{PV}}{1000} \quad (1)$$

The results for the power produced by the on-house and off-grid heat panels, and the on-house and off-grid PV panels can be seen below:

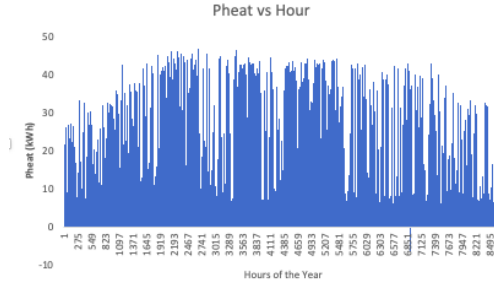


Figure 3. On-House Heat Panels where 36 panels were used and a total annual production of 48.7 MWh.

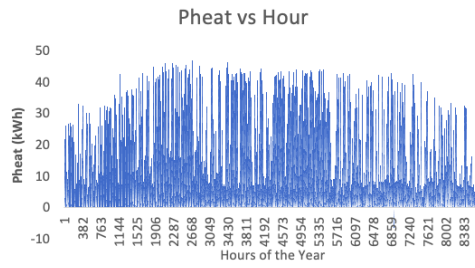


Figure 4. Off-Grid Heat Panels where 44 panels were used and a total annual production of 59.5 MWh.

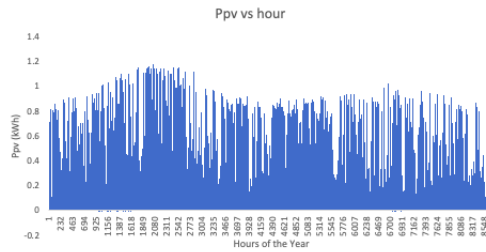


Figure 5. On-House PV Panels where 2 panels were used and a total annual production of 1.6 MWh.

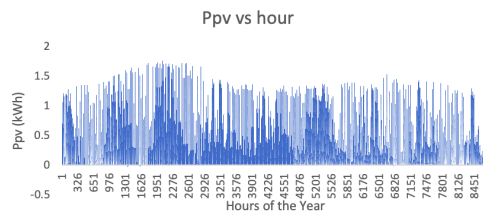


Figure 6. Off-Grid PV Panels where 3 panels were used and a total annual production of 1.6 MWh.

Therefore, the total energy produced in a year by both on-house and off-grid heat and PV panels was found to be 111.36 MWh.

5.2 Turbine Analysis

Similar to the solar panels, the total energy produced by the turbines chosen was found to be 45 kWh in Section 4.2. However, like the solar panels, this is only the case when the turbines are operating in ideal conditions as well. Therefore, this section will cover the analysis done to find the actual power produced by the wind turbines given the 2014 meteorological data.

To start the analysis, an Excel spreadsheet was used to list every hour, day, and month for a year. Then, using the analysis process taught in MECH 4692, the velocity for every hour of the year was calculated with both the Weibull and Rayleigh approximations. For both approximations, the velocity was found using a probability analysis with a Bin# of 18. This Bin# was used to break up the total wind velocity of 7 m/s, into 18 segments, where the average velocity for each was then found. Once these were found, the frequency of occurrence per month was calculated alongside the cumulative frequency of occurrence per month. The complete analysis table can be seen in Appendix E where the histogram of the values obtained can be seen below in Figure 7:

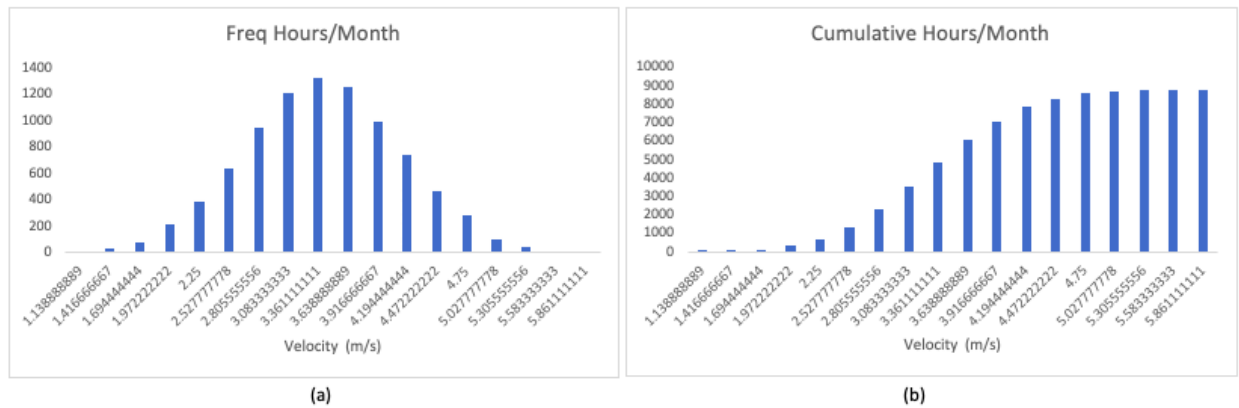


Figure 7. Frequency of occurrence per month (a) and Cumulative frequency of occurrence per month (b) for the wind velocity approximation for a given year.

To verify whether the data above was correct, the probability of each frequency was found (fi) and summed (Fi), and if it equaled 1, then the values were satisfied. A histogram of the values obtained in Appendix E can be seen below in Figure 8:

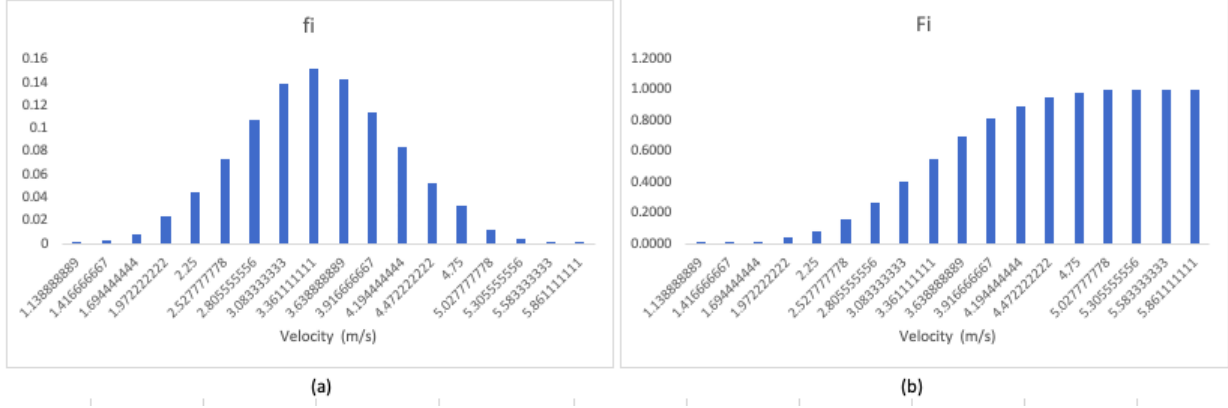


Figure 8. Wind probability, showing both the individual probability, f_i , (a) and the summed probability, F_i , (b). Additionally, this data shows the values are satisfied as they converge to 1.0.

Once the probability was found and the velocity for every month was approximated, a Vlookup was used to determine the velocity for every hour of the year. Finally, this data was then used to calculate the power for every hour of the year using Equation 2.

$$P = \frac{1}{2} \rho_{air} A_{sweep} V^3 \quad (2)$$

Where ρ_{air} is the density of air in kg/m^3 , A_{sweep} is the sweep area in m^2 , V is the velocity in m/s, and P is the power in Watts.

Once the turbine power was found for every hour of the year, the following histogram in Figure 9 was produced:

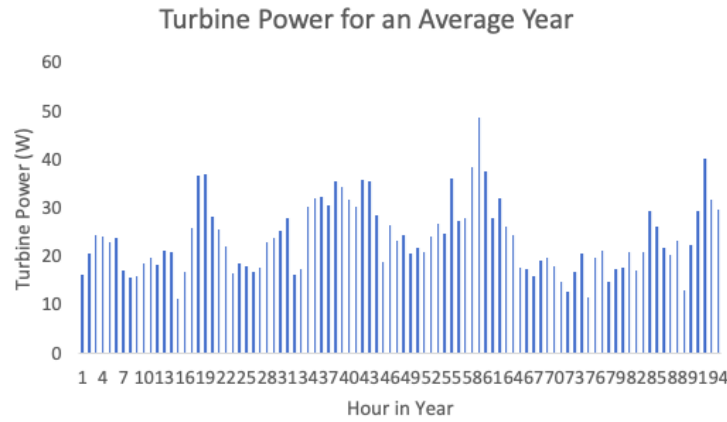


Figure 9. Turbine power for an average year showing the fluctuation of power generation where a total power of 178 kWh was produced.

5.3 Biomass Analysis

Contrary to the solar panels and turbines, the biomass analysis consisted of simply choosing the wood that will be used as the fuel source for the boiler and barbeque. This wood type was chosen to be oak as it is the most available in Winnipeg Manitoba. Additionally, the overall CO₂ and H₂O emissions were found to be very small and were assumed to be zero. Therefore, it proved that there are no GHG emissions for the biomass additions. This data can be seen in Appendix F.

6 Power and Heat Balance

After all the additions, retrofits, and supporting calculations were made, a power and heat balance was created using an Excel spreadsheet. Figure 10 shows the various power generation and usages on a plot, alongside also displaying the amount of power that is extra and will be sold back to Manitoba Hydro. The data for this analysis can be seen in Appendix G.

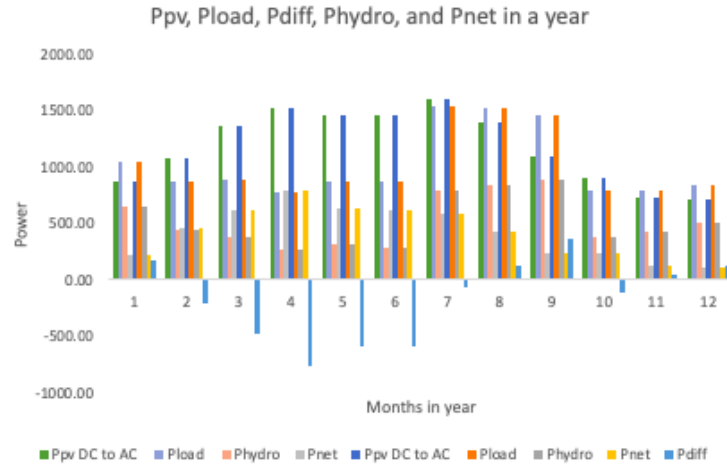


Figure 10. Various power generation and usages. Also, the amount of power that is excess which will be sold back to Manitoba Hydro is shown.

Additionally, after doing a cost savings test for 30 years, the total savings was found to be \$20,026.20 using [21], [22]. The full analysis can be found in Appendix H.

Finally, with all the changes implemented, a final RER value of 100% was achieved as there is no natural gas, gasoline, or propane being used. This data can be seen in Appendix I.

7 References

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8 Appendix A

TABLE V. Recorded data from a local hydro meter in Winnipeg, Manitoba showing the meter reading, load, and load probability.

Time (hr)	Meter Reading (kWh)	Load (kWh)	Load Probability ($f(i)$)
0	132800	0	0.00000
1	132800	0.3	0.00935
2	132800	0.5	0.01558
3	132805	1	0.03115
4	132805	1	0.03115
5	132805	1.1	0.03427
6	132805	1.2	0.03738
7	132805	1.1	0.03427
8	132810	0.9	0.02804
9	132810	0.7	0.02181
10	132810	1.6	0.04984
11	132810	1.3	0.04050
12	132815	1.5	0.04673
13	132815	2.7	0.08411
14	132820	2.3	0.07165
15	132820	2.8	0.08723
16	132820	2.4	0.07477
17	132830	2.3	0.07165
18	132830	2.1	0.06542
19	132835	1.4	0.04361
20	132835	1.8	0.05607
21	132835	1.2	0.03738
22	132840	0.5	0.01558
23	132840	0.4	0.01246
	Sum	32.1	1

9 Appendix B

Car Gasoline Emissions																	
Car	Gas Tank Size (L)	Gas Tank Size (m3)	Usage (per week)	Cost/week	January	February	March	April	May	June	July	August	September	October	November	December	Total
2015 Ford Escape	58.67388	0.05867388	0.17602164	\$ 80.00	3.34441116	2.6403246	2.6403246	1.7602164	1.4081731	1.5841948	2.6403246	3.5204328	3.3444112	2.6403246	2.6403246	2.8163462	30.979809
2012 Ford Platinum	136.275	0.136275	0.408825	\$ 150.00	4.08825	4.497075	4.08825	3.2706	3.2706	4.497075	4.497075	4.497075	4.08825	3.2706	4.497075	4.08825	48.650175
2005 Chrysler Grand Caravan	75.7082	0.0757082	0.1514164	\$ 80.00	3.3311608	3.028328	2.4226624	2.7254952	3.028328	3.3311608	1.8169968	2.8769116	3.3311608	3.4825772	3.3311608	3.1797444	35.885687

Figure 11. Annual car usages for a 2015 Ford Escape, 2012 Ford Platinum, 2005 Chrysler Grand Caravan

Month	Natural Gas Bill			Hydro Billings		Lawnmower Bill (Gasoline)		Propane Bill (20 lb/7.7 kg tank)	
	(kWh)	(m3)	Cost	(kWh)	Cost	m3	Load (kWh)	kg	Load (kWh)
January	8580.70	758.319	327.06	1393.1	75.3	0.01	109.41	0.00	0.00
February	5826.00	514.873	211.34	1113.1	70.1	0.01	109.41	0.00	0.00
March	6459.26	570.837	217.01	1433.1	77.6	0.00	0.00	0.00	0.00
April	3419.60	302.207	124.58	1423.1	79.2	0.00	0.00	0.00	0.00
May	3654.74	322.988	144.65	2563.1	95.3	0.01	109.41	1.28	17.91
June	1234.85	109.13	62.40	4513.1	125.2	0.01	109.41	1.28	17.91
July	823.24	72.754	62.22	4463.1	115.3	0.01	109.41	1.28	17.91
August	443.28	39.175	38.50	4663.1	150.6	0.01	109.41	1.28	17.91
September	664.93	58.763	39.80	1393.1	121.7	0.01	109.41	1.28	17.91
October	1741.47	153.902	78.63	1113.1	151.2	0.01	109.41	1.28	17.91
November	3641.26	321.796	163.40	833.1	130.1	0.00	0.00	0.00	0.00
December	5224.41	461.707	239.57	553.1	126.5	0.01	109.41	0.00	0.00

Figure 12. Annual energy usages for a house

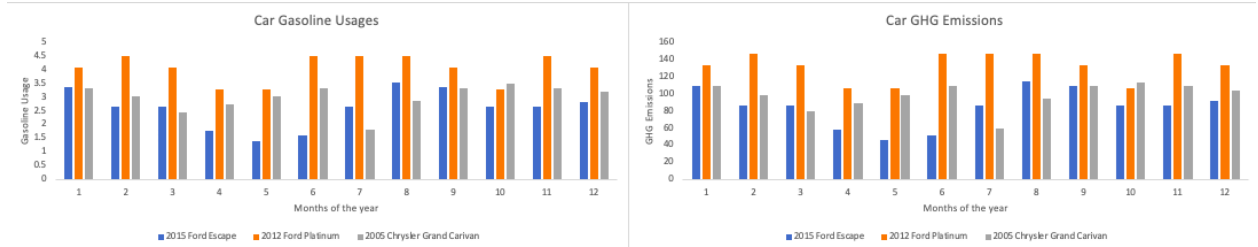


Figure 13. Annual GHG emissions and gasoline usages for a 2015 Ford Escape, 2012 Ford Platinum, 2005 Chrysler Grand Caravan

10 Appendix C

TABLE VI. Conversion values for natural gas, gasoline, and propane.

	Natural Gas	Gasoline	Propane
Density (kg/m3)	0.78	744.7	507.21
HHV (MJ/kg)	52.225	46.536	50.235
Usage	7.7 kg tank per year	2.5 gal/month	
kg CO ₂ /gallon	0.0024	0.1235	0.173913043
kg CO ₂ /m3	0.6436	32.6138	45.94298728

11 Appendix D

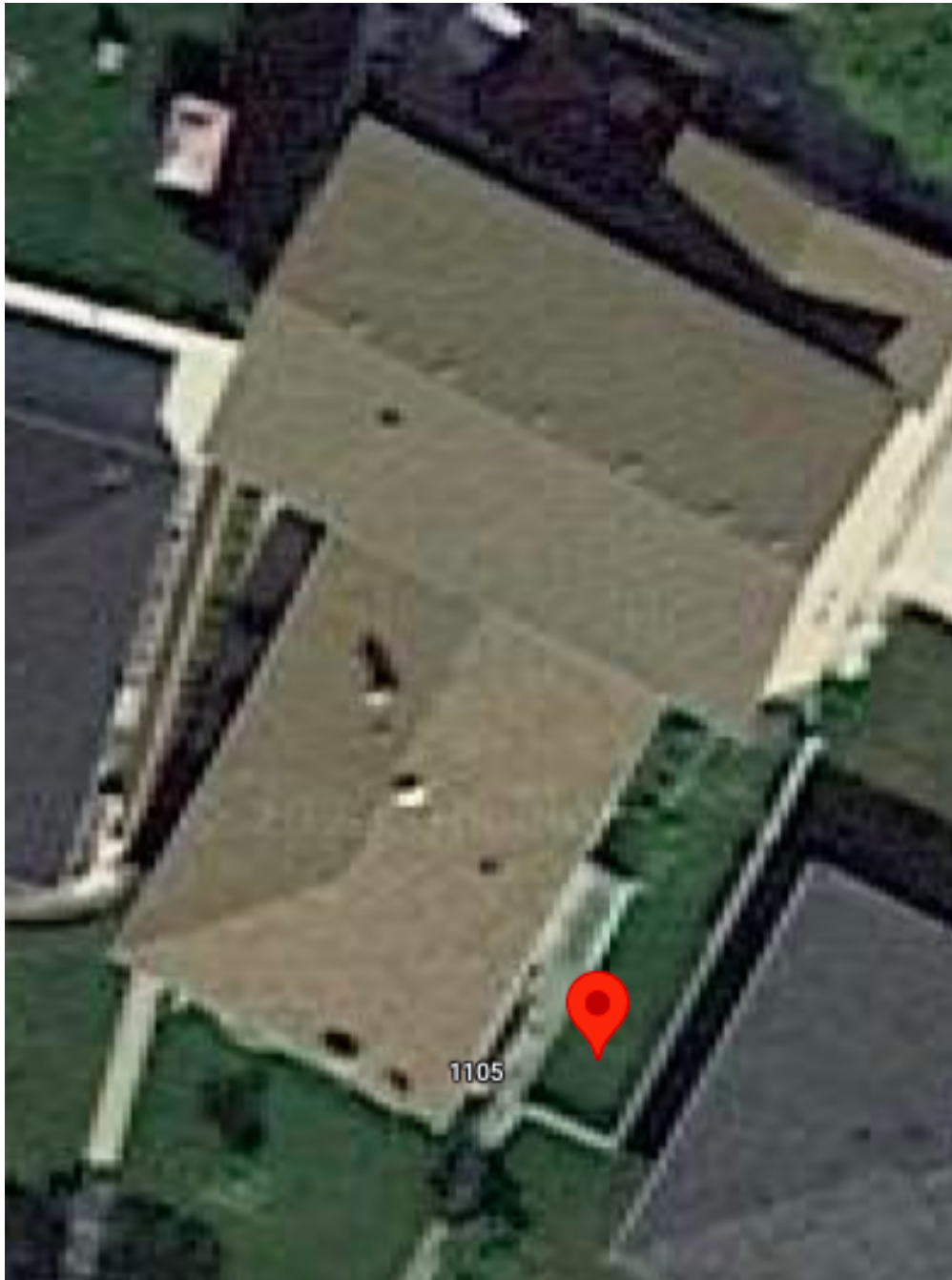


Figure 14. Current house

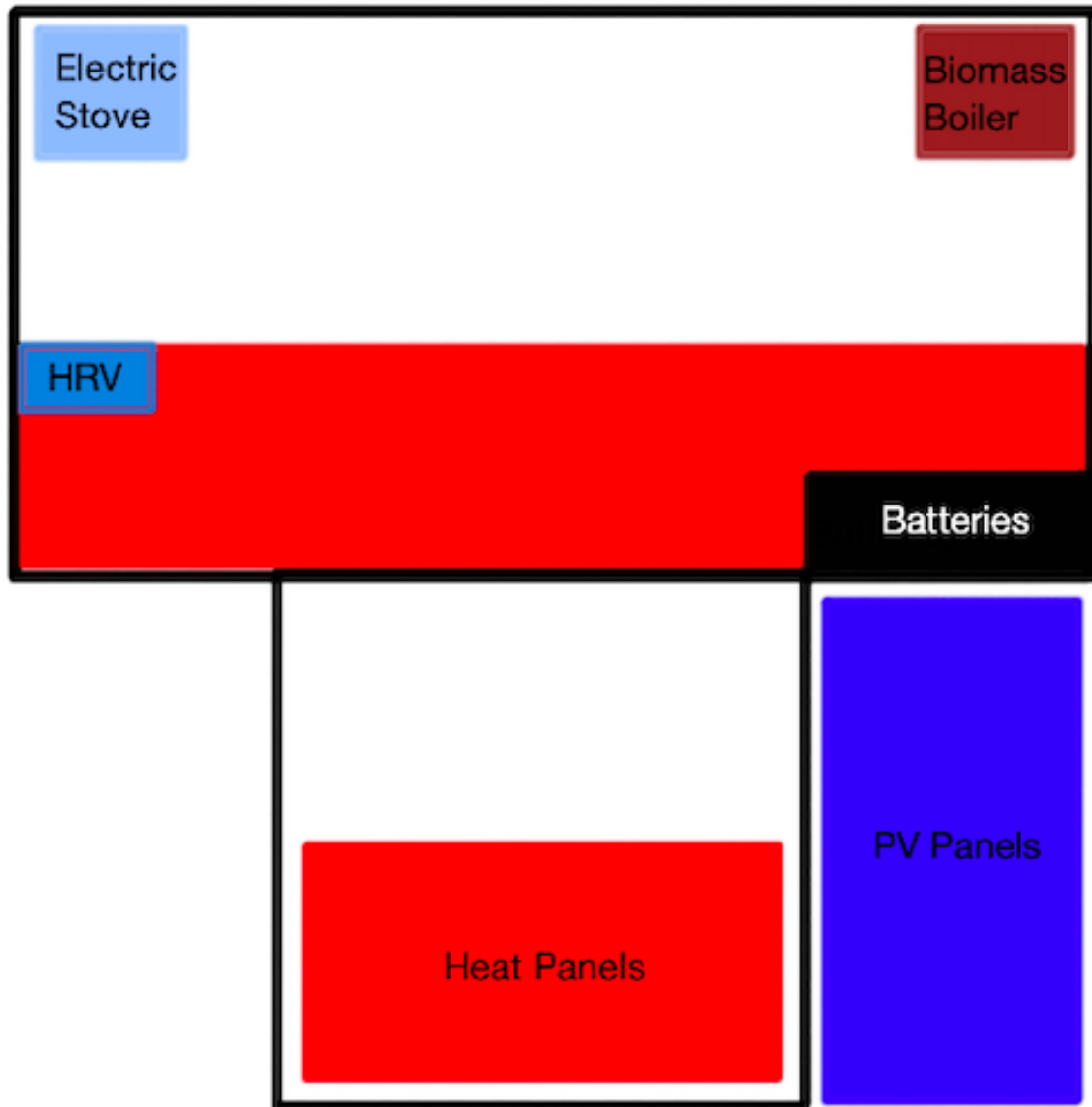


Figure 15. Concept house

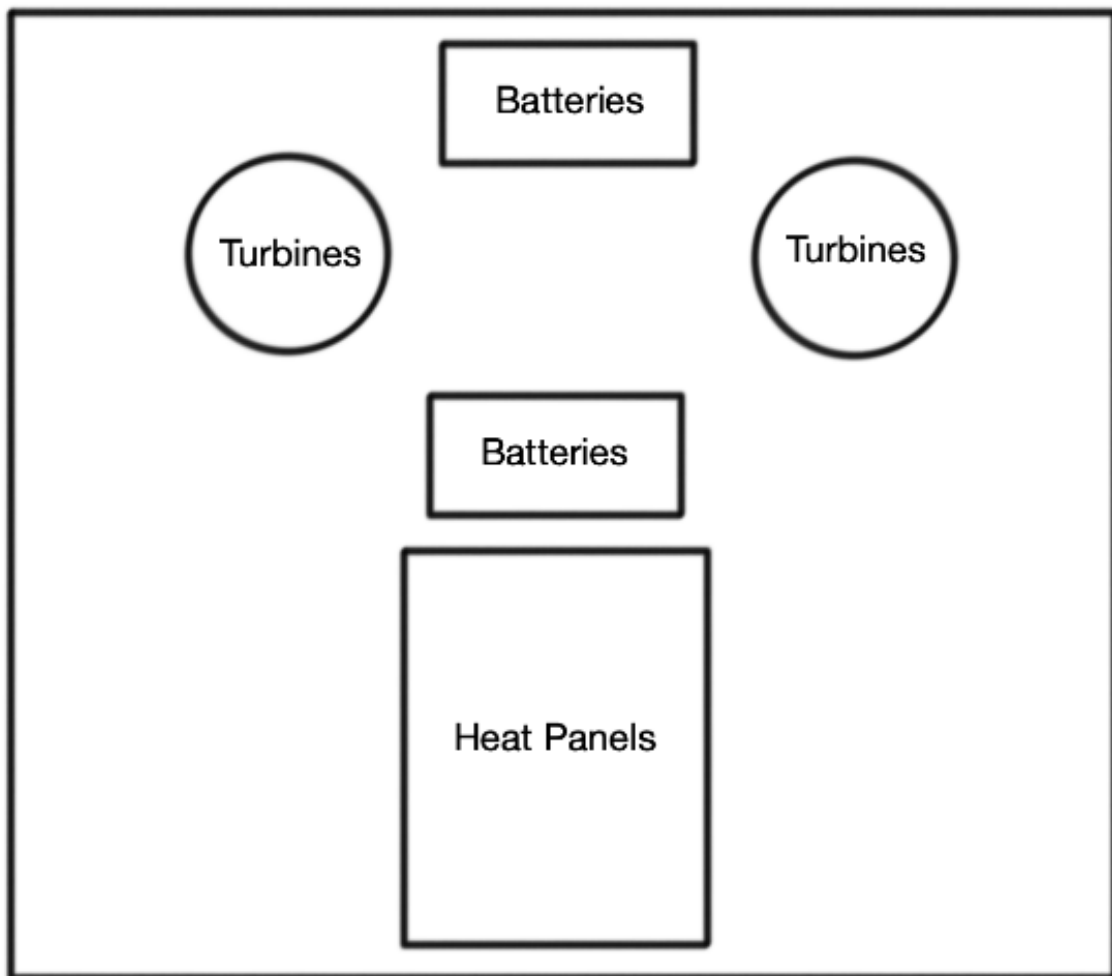


Figure 16. Off Grid Concept

12 Appendix E

Bin #	Vel	Avg	Freq Hours/Month	Cumulative Hours/Month	f_i	Fi	$f_i V_i$	$f_i (V_i - V_m)^2$
1	2.278	1.13888889	5	5	0.0006	0.0006	0.001	0.003
2	2.556	1.41666667	30	35	0.0034	0.0040	0.005	0.014
3	2.833	1.69444444	79	114	0.0090	0.0130	0.015	0.026
4	3.111	1.97222222	216	330	0.0247	0.0377	0.049	0.051
5	3.389	2.25	389	719	0.0444	0.0821	0.100	0.059
6	3.667	2.52777778	641	1360	0.0732	0.1553	0.185	0.056
7	3.944	2.80555556	948	2308	0.1082	0.2635	0.304	0.039
8	4.222	3.08333333	1217	3525	0.1389	0.4024	0.428	0.014
9	4.500	3.36111111	1330	4855	0.1518	0.5543	0.510	0.000
10	4.778	3.63888889	1256	6111	0.1434	0.6977	0.522	0.008
11	5.056	3.91666667	996	7107	0.1137	0.8114	0.445	0.030
12	5.333	4.19444444	741	7848	0.0846	0.8960	0.355	0.053
13	5.611	4.47222222	463	8311	0.0529	0.9489	0.236	0.060
14	5.889	4.75	290	8601	0.0331	0.9820	0.157	0.060
15	6.167	5.02777778	103	8704	0.0118	0.9937	0.059	0.031
16	6.444	5.30555556	43	8747	0.0049	0.9986	0.026	0.018
17	6.722	5.58333333	7	8754	0.0008	0.9994	0.004	0.004
18	7.000	5.86111111	5	8759	0.0006	1.0000	0.003	0.003
		3.5	8759		1.0000		3.41	0.73
					sum f_i		V_m	σ_i

Figure 17. Bin# Calculations to find the probability curve for the wind turbine

Rayleigh	$f(V)$	$F(V)$
	0.21711528	0.82718364
	0.22242067	0.75207545
	0.22282263	0.67844238
	0.21878118	0.60977041
	0.21088495	0.54754945
	0.19980876	0.49209725
	0.1862708	0.44312107
	0.17099243	0.40005297
	0.15466291	0.36223468
	0.13791087	0.32901245
	0.12128339	0.29978133
	0.10523303	0.27400213
	0.09011231	0.25120435
	0.07617484	0.23098238
	0.06358176	0.2129889
	0.05241222	0.19692748
	0.04267621	0.18254541
	0.03432875	0.16962718

Figure 18. Rayleigh approximation data

Weibull	k	c	f(V)	F(V)
	5.34697936	4.90078517	64.543574	0.01648786
			34.3957505	0.03029125
			19.3668444	0.05200495
			11.3455754	0.08429299
			6.82428678	0.12987075
			4.16357281	0.19102565
			2.54632246	0.26893112
			1.54231514	0.36283883
			0.9136014	0.46937039
			0.52212055	0.58226699
			0.28365917	0.69298794
			0.14414063	0.79233922
			0.06729236	0.87281513
			0.02829744	0.93074989
			0.01048685	0.96716406
			0.00334345	0.98675006
			0.00089305	0.99556504
			0.0001941	0.99880307

Figure 19. Weibull approximation data

13 Appendix F

	Cp	Moles	Molar Mass	Mass	Energy
CO2	1.2006843 kJ/kg°C	7.2747E-06 moles	0.04401 kg/mol	3.2016E-07 kg	1.05713E-07 MJ
O2	1.06643146 kJ/kg°C	3.5828E-06 moles	0.031999 kg/mol	1.1465E-07 kg	3.36219E-08 MJ
N2	1.15626276 kJ/kg°C	4.246E-05 moles	0.0280134 kg/mol	1.1895E-06 kg	3.78215E-07 MJ
H2O	2.2271773 kJ/kg°C	8.9144E-06 moles	0.01801528 kg/mol	1.6059E-07 kg	9.83601E-08 MJ
Delta T	275 °C				

Figure 20. Biomass data showing the overall emissions is very small and therefore can be assumed to be zero.

14 Appendix G

Month	Ppv DC to AC	Pload	Pdiff (kWh)	Phydro (kWh)	Pnet (kWh)	Pnet over	Total Cost 15th yr(\$)
January	877.71	1052.74	175.03	650.58	223.45	0.00	\$62.05
February	1086.95	872.74	-214.21	452.12	465.18	13.05	\$0.00
March	1360.23	884.54	-475.70	376.15	623.45	247.30	\$0.00
April	1527.16	770.94	-756.22	264.33	798.53	534.20	\$0.00
May	1465.87	877.09	-588.78	313.26	642.96	329.69	\$0.00
June	1453.36	871.44	-581.92	291.26	625.86	334.60	\$0.00
July	1609.38	1541.44	-67.94	788.09	580.84	0.00	\$30.11
August	1392.09	1521.44	129.35	849.22	436.21	0.00	\$60.00
September	1095.60	1461.44	365.84	895.69	240.07	0.00	\$95.24
October	912.20	797.09	-115.11	382.36	241.55	0.00	\$20.45
November	737.59	786.49	48.90	429.33	120.34	0.00	\$44.88
December	719.85	840.47	120.62	503.24	114.23	0.00	\$56.51

Figure 21. Power and heat balance showing the leftover power after usages per month and the total amount of money that is needed to be paid to Manitoba Hydro.

15 Appendix H

Electricity Price Current	0.09324
Electricity Price 15 years	0.145264882
Total Electricity Cost 30 years with renewables	\$ 11,077.06
Total Electricity Cost 30 years with no renewables	\$ 53,506.28
Total Savings over 30 years	\$ 42,429.21
Biomass Price Current	\$213.72
Biomass Price 15 years	\$248.12
Total Heating Cost 30 years with renewables	\$ 39,502.25
Total Heating Current Yearly Cost	\$ 1,709.16
Total Heating Cost 30 years with no renewables	\$ 59,528.45
Total Savings over 30 years	\$ 20,026.20

Figure 22. Power and heat balance showing the total amount saved in 30 years if all renewables were implemented.

16 Appendix I

Total Load Before Retrofits	1200 MWh
Turbine	0.18 MWh
On-House Heat Panels	48.69 MWh
On-House PV Panels	1.58 MWh
Off-Grid Heat Panels	59.51 MWh
Off-Grid PV Panels	1.58 MWh
Total Power Produced	111.54 MWh
Natural Gas	0 MWh
Propane	0 MWh
Total Load After Retrofits	0.04 MWh
Car Loads	9.06 MWh
Hydro	49.80 MWh
Total Power Needed	58.90 MWh
Excess Power	52.63 kWh
RER	100 %

Figure 23. Final RER analysis